PR Summary: Reduce Agentic Codexa Overhead

- Project: codexa
- Branch: codex/general/codexa-20260718-000156-focus-orientation
- Base: main
- Source head before summary refresh: 81ef637f913e
- Subject: perf(mcp): reduce agentic Codexa overhead

Goal and Outcome

This change targets the archived agent A/B failure in which the Codexa-enabled arm used 5.94x the control’s input tokens while producing the same verified completion result. The current candidate removes the mandatory lifecycle call chain, defaults agent hosts to a three-tool MCP surface, and tells agents to use zero Codexa calls for exact local tasks.

An authenticated one-pair regression smoke on the same checked-in task did not reproduce the 5.94x failure:

Actual model usage	Control	Current Codexa treatment	Ratio
Input tokens	108,683	127,190	1.17x
Cached input tokens	76,032	109,568	1.44x
Output tokens	3,065	3,551	1.16x
Codexa tool calls	0	0	no difference

The treatment used the intended zero-call route, passed the public tests, and passed all committed hidden behavior cases plus 40 generated cases. The single control run passed its public tests but failed the leading/trailing C1 control case. This is a non-confirmatory one-pair smoke, not a broad product-effect estimate. It has no task-clustered interval or comparable provider-cost metric.

What Changed

Smaller default agent surface

- Agent launchers now default to the core MCP profile: search, changeplan, and capabilities.
- capabilities retains access to all 22 logical operations, while the full direct-tool profile remains an explicit opt-in.
- Server instructions and tool guidance make raw source reads terminal when an exact target is already known.

Selective lifecycle cadence

- Exact file, symbol, command, config, and quoted-error work can use zero Codexa calls.
- A normal bounded change usually uses at most two calls; only an ambiguous, materially risky change without a managed completion gate uses the narrow three-call exception.
- Managed hooks coalesce duplicate work and retain an evidence-bearing final review only when the host can provide meaningful verification evidence.

Bounded transport results

- Automatic and concise results stay inside a hard serialized-result budget.
- Detailed evidence can be fetched through content-addressed, repository-bound MCP resources instead of being repeated inline.
- Repeated equivalent calls return bounded unchanged receipts while preserving the same logical operation contract.

Routing and lifecycle hardening

- Exact target authority, source dependencies, mutation intent, and directed task language are kept consistent across search, planning, and review.
- Snapshot governance preserves the last durable plan across interrupted replans without colliding with dotted task IDs.
- Rollback artifacts use an isolated internal namespace with symlink and containment checks, preventing writes outside the repository.
- The storage helpers were extracted to keep task-snapshots.ts below the repository’s 1,000-line source-hygiene ceiling.

Verification

Gate	Result
npm run check	72 test files; 900 passed; 1 intentional skip
Claude integration smokes	28 command checks and 89 hook checks passed
Package hygiene	Generated npm and plugin contents passed
Installed-package smoke	31 checks passed against the packed 0.15.0 tarball
Source hygiene	Passed; task-snapshots.ts is 994 lines
Adversarial review	Latest strict pass: NO ACTIONABLE FINDINGS
Bloat audit	No high-confidence removable production bloat found

The final adversarial bar was limited to core regressions, security or write authority, crash or data loss, and direct redundant-agent-call regressions. Actionable findings were fixed in separate Conventional Commits; speculative polish and refactor suggestions were not accepted.

Transport Evidence

The deterministic benchmark measures decoded JSON application bytes, not model tokens or provider cost. On the final clean candidate checkout:

Same-build full vs. core	Full	Core	Reduction
tools/list	68,196 B	8,745 B	87.2%
Startup + advertisement + discovery	88,032 B	26,622 B	69.8%
Advertised logical operations	22	22	no loss

Against the pinned v0.12.0 source build, startup plus advertisement and discovery fell from 60,409 B to 26,622 B (55.9%), the first task result fell from 55,285 B to 13,852 B (74.9%), and the repeated-result median fell from 55,285 B to 6,117 B (88.9%).

The hot-path benchmark also passed every threshold. Representative p95 values were 4,707 ms for indexing, 496 ms for status, 223 ms for MCP freshness, 30 ms for MCP repository map, and 118 ms for an explicit-file MCP task brief.

Limits and Rollback

- The actual-token result is one task and one pair. A registered, held-out, multi-task experiment is still required before claiming general agent value.
- Task-result payloads are not universally smaller in same-build comparisons; the largest deterministic win is exposure and repeated-result overhead.
- The full direct-tool profile remains available for hosts that explicitly need it.
- If the selective defaults cause a regression, revert this PR; no persistence migration or external service rollback is required.